
CE 401: Design of Steel Structures

Course Code	CE 401
Course Name	Design of Steel Structures
L-T-P-C	2-1-0-3
Prerequisites	CE 301- Strength of Materials and Structures
Intended for	UG
Distribution	Discipline Core
Approval	9th Senate (May 2016)

Course Contents

- **Design of connections**
Rivet, bolt and welding [5 Lectures]
- **Tension and Compression member**
load calculation and design [5 Lectures]
- **Module III**
Design of members subjected to Unsymmetrical Bending [5 Lectures]
- **Module IV**
Design of Plate Girder. [5 Lectures]
- **Module V** Design of Base-plate and Anchor [5 Lectures]
- **Module VI**
Design of Composite beams and Slabs [5 Lectures]
- **Module VII**
Plastic analysis and design of continuous beam and simple frame . [5 Lectures]
- **Module VIII**
Design of a multi-storey building and industrial shed. [5 Lectures]
- **Module IX**
An overview of design for fatigue. [5 Lectures]

Text Books

1. N. Subramanian **Design of steel structures**, Oxford University Press, 2008.
2. S. K. Duggal, **Limit state design of steel structures**, McGraw Hill Education, 2014.

References

1. C.G. Salmon, J.E. Johnson, and F.A. Malhas, **Steel structures: design and behavior: emphasizing load and resistance factor design**, 5th Edition, Pearson, 2008.
2. Trahair, N.S., M. A. Bradford, D. Nethercot and L. Gardner., **The behavior and design of steel structures to EC3**, Taylor and Francis, 2008.
3. A. Nussbaumer, L. , Borges and L. Davaine, **Fatigue Design of Steel and Composite Structures**, John Wiley & Sons, 2011.